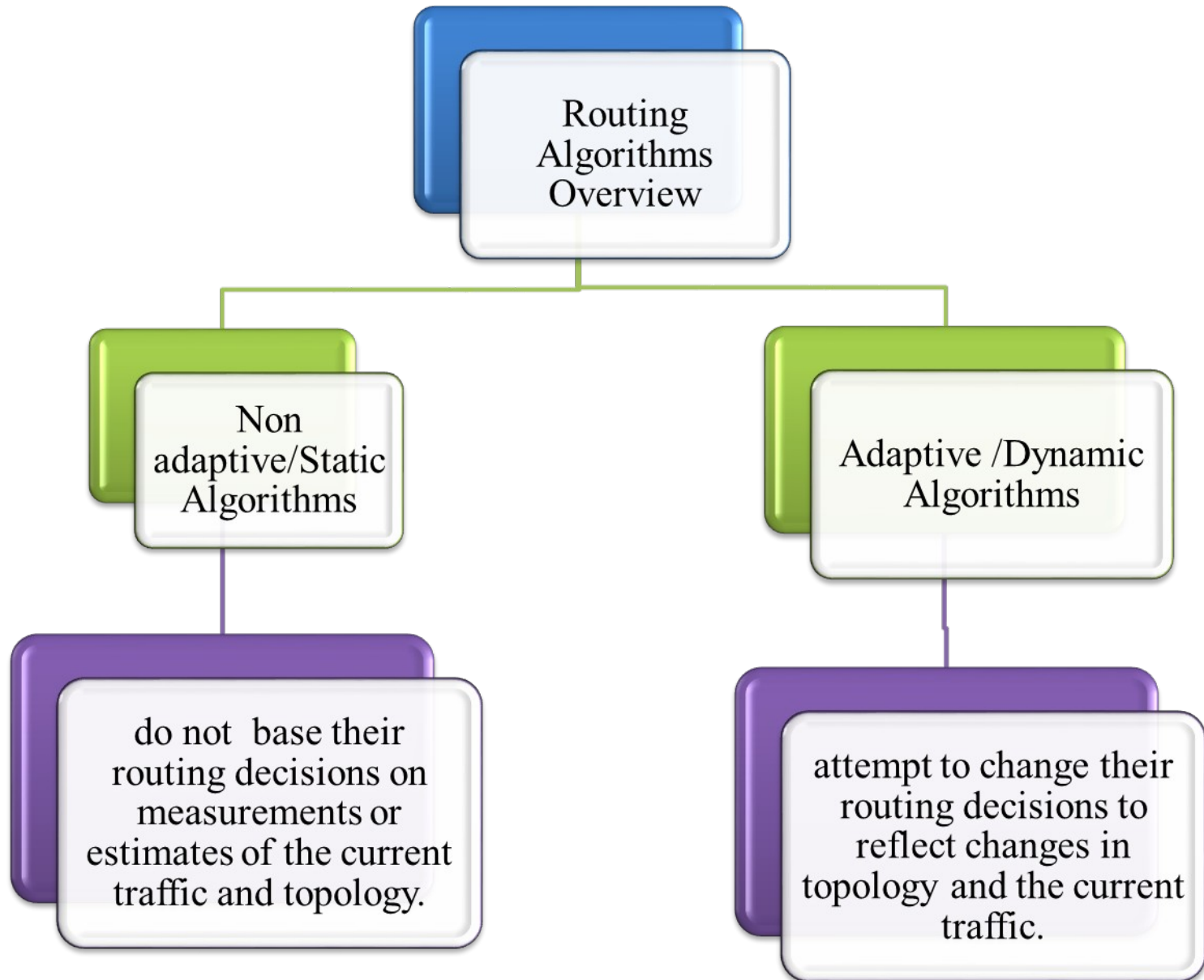


NETWORK LAYER

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Routing Algorithms



Routing protocols

Intradomain

Interdomain

Distance
vector

RIP

Link
state

OSPF

Path
vector

BGP

Distance Vector / Link State Routing

Distance Vector Routing

- It is a dynamic routing algorithm in which each router computes distance between itself and each possible destination i.e. its immediate neighbors.
- The router share its knowledge about the whole network to its neighbors and accordingly updates table based on its neighbors.
- The sharing of information with the neighbors takes place at regular intervals.
- It makes use of Bellman Ford Algorithm for making routing tables.
- Problems
 - Count to infinity problem
 - Good news spread fast and bad news spread slowly.
 - Persistent looping problem

Link State Routing

- It is a dynamic routing algorithm in which each router shares knowledge of its neighbors with every other router in the network.
- A router sends its information about its neighbors only to all the routers through flooding.
- Information sharing takes place only whenever there is a change.
- It makes use of Dijkstra's Algorithm for making routing tables.
- Problems
 - Heavy traffic due to flooding of packets.
 - Flooding can result in infinite looping which can be solved by using Time to leave (TTL) field.

Distance vector routing

- In distance vector routing, the least cost route between any two nodes is the route with minimum distance (hop count).
- In this protocol each node maintains a vector (table) of minimum distances to every node
- Three phases to built a routing table - Initialization , Sharing, Updating
- In distance vector routing, the least-cost route between any two nodes is the route with minimum distance.
- In this protocol, as the name implies, each node maintains a vector (table) of minimum distances to every node.
- The table at each node also guides the packets to the desired node by showing the next stop in the route (next-hop routing).

Initialization of tables in distance vector routing

In distance vector routing, each node shares its routing table with its immediate neighbors periodically and when there is a change.

To Cost Next

A	0	—
B	5	—
C	2	—
D	3	—
E	∞	—

A's Table

To Cost Next

A	3	—
B	∞	—
C	∞	—
D	0	—
E	∞	—

D's Table

To Cost Next

A	2	—
B	4	—
C	0	—
D	∞	—
E	4	—

C's Table

To Cost Next

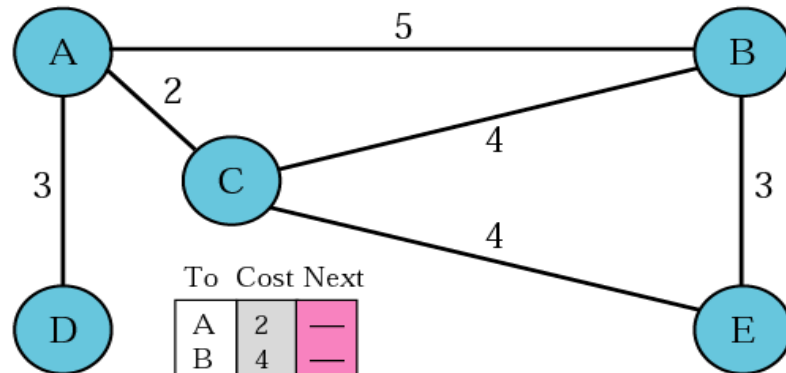
A	5	—
B	0	—
C	4	—
D	∞	—
E	3	—

B's Table

To Cost Next

A	∞	—
B	3	B
C	4	C
D	∞	—
E	0	D

E's Table



To Cost

A	2
B	4
C	0
D	∞
E	4

Received from C

To Cost Next

A	4	C
B	6	C
C	2	C
D	∞	C
E	6	C

A's Modified Table

Compare

To Cost Next

A	0	—
B	5	—
C	2	—
D	3	—
E	∞	—

A's Old Table

Updating in distance vector routing

To Cost Next

A	0	—
B	5	—
C	2	—
D	3	—
E	6	C

A's New Table

Link State Routing

Link state routing has a different philosophy from that of distance vector routing. In link state routing, if each node in the domain has the entire topology of the domain the list of nodes and links, how they are connected including the type, cost (metric), and condition of the links (up or down)-the node can use Dijkstra's algorithm to build a routing table.

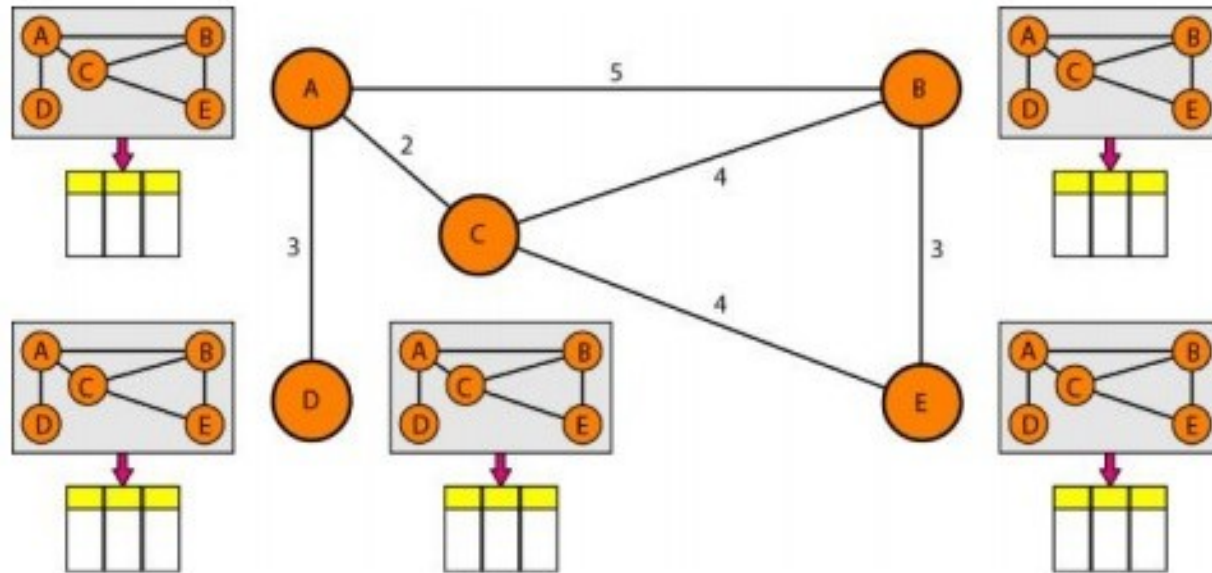


Figure 3.50 Concept of link state routing

The figure shows a simple domain with five nodes. Each node uses the same topology to create a routing table, but the routing table for each node is unique because the calculations are based on different interpretations of the topology. This is analogous to a city map. While each person may have the same map, each needs to take a different route to reach her specific destination.

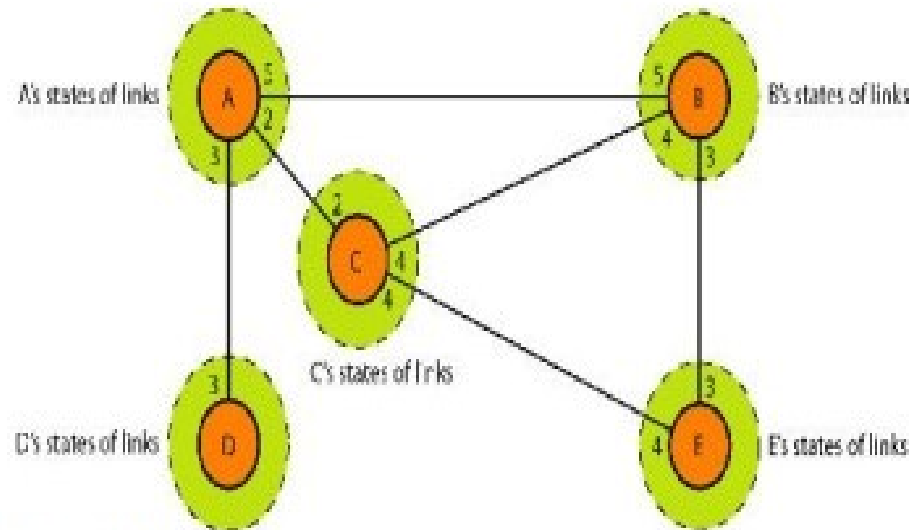


Figure 3.51 Link state knowledge

RIP(Routing Information Protocol)

RIP is an intra-domain routing protocol used within an autonomous system. Here, intra-domain means routing the packets in a defined domain, for example, web browsing within an institutional area. To understand the RIP protocol.

following points explains how RIP works.

- RIP is based on the distance vector-based strategy, so we consider the entire structure as a graph where nodes are the routers, and the links are the networks.
- In a routing table, the first column is the destination, or we can say that it is a network address.
- The cost metric is the number of hops to reach the destination. The number of hops available in a network would be the cost. The hop count is the number of networks required to reach the destination.

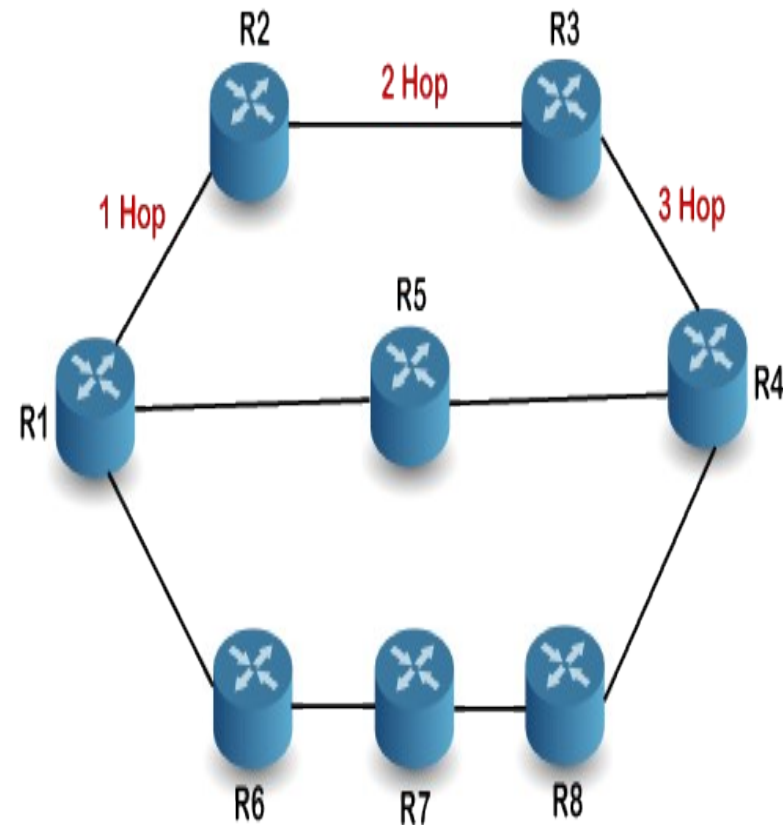
➤ In RIP, infinity is defined as 16, which means that the RIP is useful for smaller networks or small autonomous systems. The maximum number of hops that RIP can contain is 15 hops, i.e., it should not have more than 15 hops as 16 is infinity.

➤ The next column contains the address of the router to which the packet is to be sent to reach the destination.

How is hop count determined?

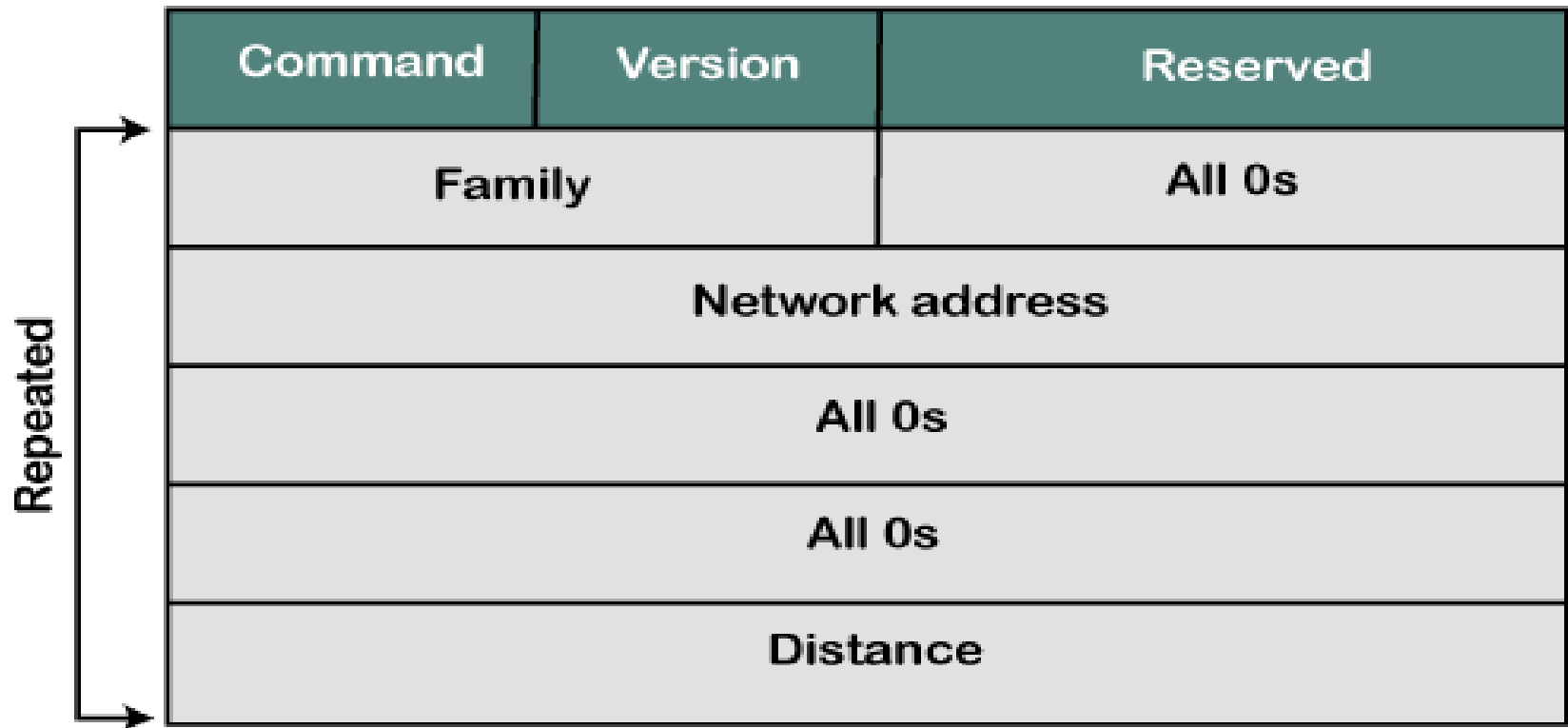
When the router sends the packet to the network segment, then it is counted as a single hop.

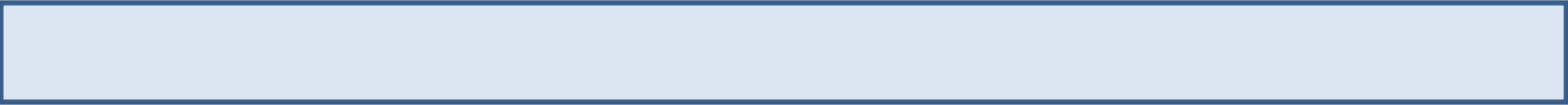
In the above figure, when the router 1 forwards the packet to the router 2 then it will count as 1 hop count. Similarly, when the router 2 forwards the packet to the router 3 then it will count as 2 hop count, and when the router 3 forwards the packet to router 4, it will count as 3 hop count. In the same way, RIP can support maximum upto 15 hops, which means that the 16 routers can be configured in a RIP.



RIP Message Format

Now, we look at the structure of the RIP message format. The message format is used to share information among different routers. The RIP contains the following fields in a message:





Command: It is an 8-bit field that is used for request or reply. The value of the request is 1, and the value of the reply is 2.

Version: Here, version means that which version of the protocol we are using. Suppose we are using the protocol of version1, then we put the 1 in this field.

Reserved: This is a reserved field, so it is filled with zeroes.

Family: It is a 16-bit field. As we are using the TCP/IP family, so we put 2 value in this field.

Network Address: It is defined as 14 bytes field. If we use the IPv4 version, then we use 4 bytes, and the other 10 bytes are all zeroes.

Distance: The distance field specifies the hop count, i.e., the number of hops used to reach the destination.

Advantages of RIP

The following are the advantages of a RIP protocol:

It is easy to configure

It has less complexity

The CPU utilization is less.

•Disadvantages of RIP.

- The RIP is a classful routing protocol, so it does not support the VLSM (Variable Length Subnet Mask).
- The classful routing protocol is a protocol that does not include the subnet mask information in the routing updates.
- It faces a problem of Slow convergence. Whenever the router or link fails, then it often takes minutes to stabilize or take an alternative route; This problem is known as Slow convergence.

Open Shortest Path First (OSPF) protocol States

Open Shortest Path First (OSPF) is a link-state routing protocol that is used to find the best path between the source and the destination router using its own Shortest Path First.

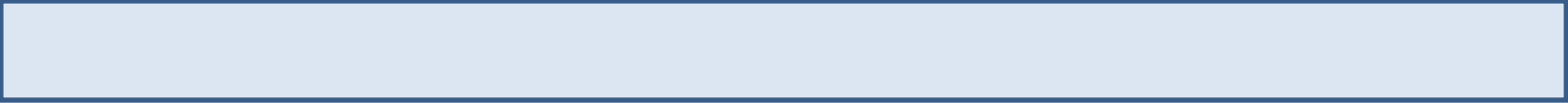
OSPF is developed by Internet Engineering Task Force (IETF) as one of the Interior Gateway Protocol (IGP), i.e., the protocol which aims at moving the packet within a large autonomous system or routing domain.

It is a network layer protocol which works on the protocol number 89 and uses AD value 110.

OSPF terms –

Router I'd – It is the highest active IP address present on the router. First, highest loopback address is considered. If no loopback is configured then the highest active IP address on the interface of the router is considered.

Router priority – It is a 8 bit value assigned to a router operating OSPF, used to elect DR and BDR in a broadcast network.



Designated Router (DR) – It is elected to minimize the number of adjacency formed. DR distributes the LSAs to all the other routers. DR is elected in a broadcast network to which all the other routers shares their DBD. In a broadcast network, router requests for an update to DR and DR will respond to that request with an update.

Backup Designated Router (BDR) – BDR is backup to DR in a broadcast network. When DR goes down, BDR becomes DR and performs its functions.

DR and BDR election – DR and BDR election takes place in broadcast network or multi-access network.

OSPF states – The device operating OSPF goes through certain states. These states are:

Down – In this state, no hello packet have been received on the interface.

Note – The Down state doesn't mean that the interface is physically down. Here, it means that OSPF adjacency process has not started yet.

INIT – In this state, hello packet have been received from the other router.

2WAY – In the 2WAY state, both the routers have received the hello packets from other routers. Bidirectional connectivity has been established.

Exstart – In this state, NULL DBD are exchanged. In this state, master and slave election take place. The router having the higher router ID becomes the master while other becomes the slave.

Exchange – In this state, the actual DBDs are exchanged.

Loading – In this state, LSR, LSU and LSA (Link State Acknowledgement) are exchanged.

Full – In this state, synchronization of all the information takes place. OSPF routing can begin only after the Full state.

OSPF header

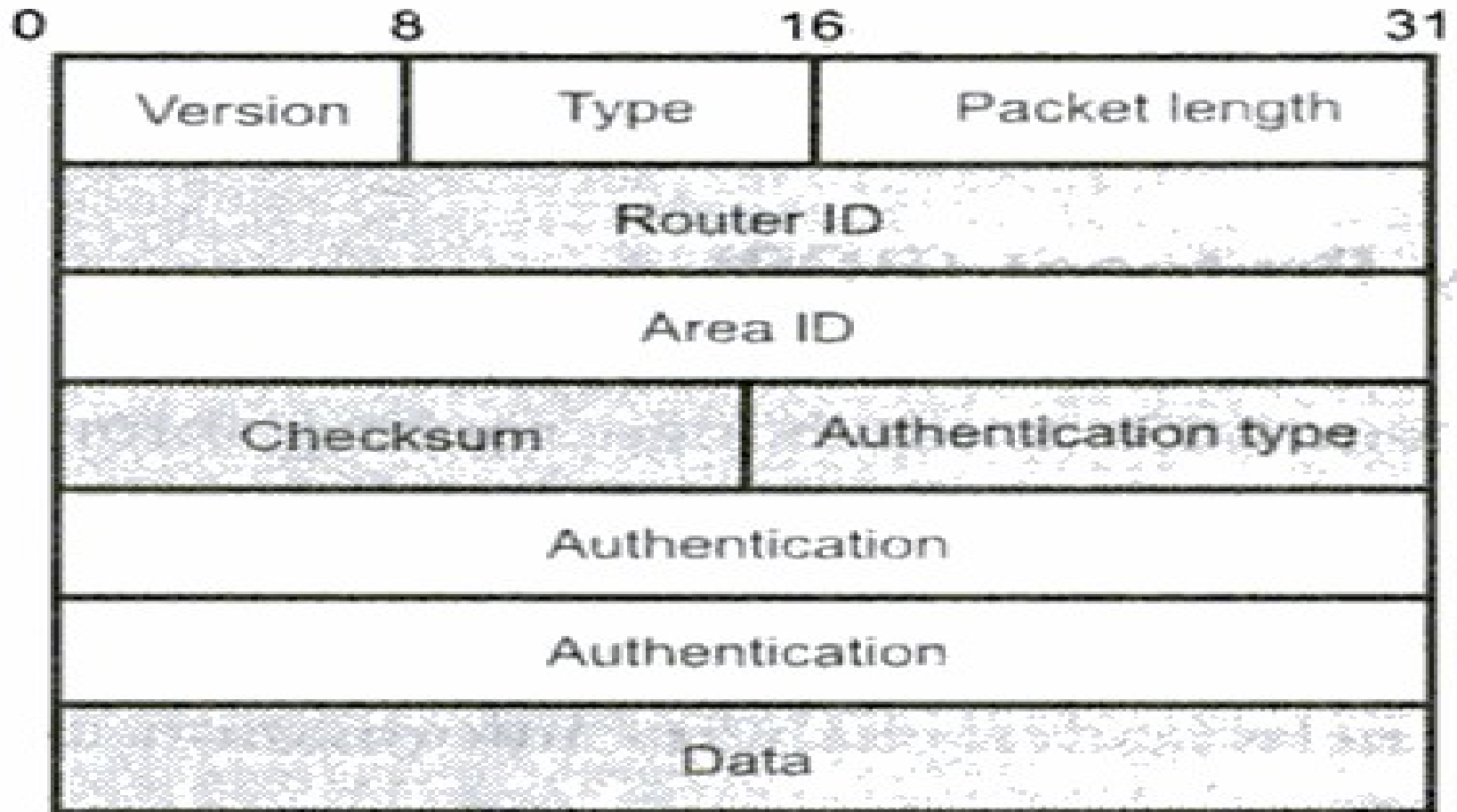


Fig.1 OSPF common header

OSPF header analysis is given below.

1. Version: This field specifies the protocol version.
2. Type: This field indicates messages as one of the following type.
 - a. Hello
 - b. Database description
 - c. Link status
 - d. Link status update
 - e. Link status acknowledgement.
3. Packet length: This field specifies the length of OSPF packet in bytes, including the OSPF header.
4. Router ID : It identify the sending router. This field is typically set to the IP address of one of its interfaces.
5. Area ID: This field identifies the area this packet belongs to (Transmitted) .
6. Checksum :The checksum field is used to detect errors in the packet. The checksum is performed on the entire packet.
7. Authentication type: It identifies the authentication type that is used.
8. Authentication : This field includes a value from the authentication type.

OSPF supports/provides/advantages –

Both IPv4 and IPv6 routed protocols

Load balancing with equal cost routes for same destination

VLSM and route summarization

Unlimited hop counts

Trigger updates for fast convergence

A loop free topology using SPF algorithm

Run on most routers

Classless protocol

RIP

RIP Stands for Routing Information Protocol.

RIP works on Bellman Ford algorithm.

It is a Distance Vector protocol and it uses the distance or hops count to determine the transmission path.

It is basically use for smaller size organization.

It allows a maximum of 15 hops.

It is not a more intelligent dynamic routing protocol.

The networks are classified as areas and tables here.

Its administrative distance is 120.

RIP is a Distance Vector Protocol.

OSPF

OSPF stands for Open Shortest Path First.

OSPF works on Dijkstra algorithm.

It is a link state protocol and it analyzes different sources like the speed, cost and path congestion while identifying the shortest path.

It is basically use for larger size organization in the network.

There is no such restriction on the hop count.

It is a more intelligent routing protocol than RIP.

The networks are classified as areas, sub areas, autonomous systems and backbone areas here.

Its administrative distance is 110.

OSPF is a Link State Protocol.

EIGRP:

➤ Enhanced Interior Gateway Routing Protocol (EIGRP) is a dynamic routing Protocol which is used to find the best path between any two layer 3 device to deliver the packet.

➤ EIGRP works on network layer Protocol of osi model and uses the protocol number 88.

➤ It uses metric to find out best path between two layer 3 device (router or layer 3 switch) operating EIGRP.

It uses some messages to communicate with the neighbor devices that operates EIGRP. These are :-

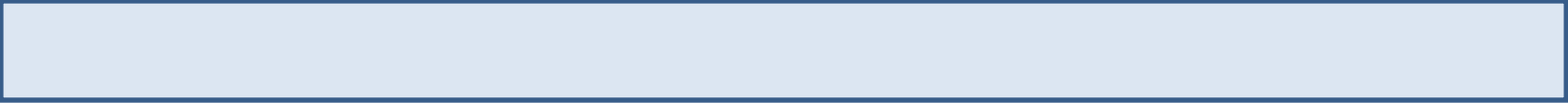
Hello message-These messages are keep alive messages which are exchanged between two devices operating EIGRP. These messages are used for neighbor discovery/recovery, if there is any device operating EIGRP or if any device(operating EIGRP) coming up again.

NULL update-It is used to calculate SRTT(Smooth Round Trip Timer) and RTO(retransmission time out)

SRTT:The time is taken by a packet to reach neighboring router and the acknowledgment of the packet to reach to the local router.

RTO: If a multicast fails then unicast are being sent to that router. RTO is the time for which the local router waits for an acknowledgment of the packet.

Full Update – After exchanging hello messages or after the neighbourhood is formed, these messages are exchanged. This message contains all the best routes.



Partial update-These messages are exchanged when there is a topology change and new links are added. It contains only the new routes, not all the routes. These messages are multicast.

Query message-These messages are multicast when the device is declared dead and it has no routes to it in its topology table.

Reply message – These messages are the acknowledgment of the query message sent to the originator of the query message stating the route to the network which has been asked in the query message.

Acknowledgement message

It is used to acknowledge EIGRP update, queries, and replies. Acks are hello packets that contain no data.

The characteristics of EIGRP follow:

- Hybrid routing protocol (distance vector that has link-state protocol characteristics).
- Uses IP protocol 88.
- Classless protocol (supports VLSMs).
- Default composite metric uses bandwidth and delay.
- You can factor load and reliability into the metric.
- Sends partial route updates only when there are changes.
- Support for authentication.

Advantages and Disadvantages:

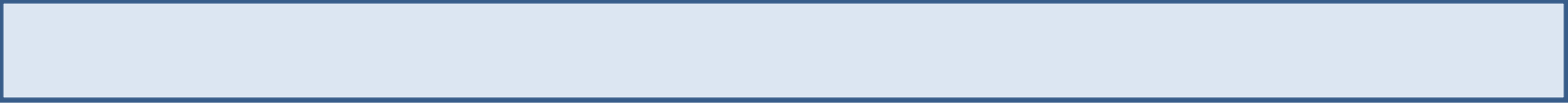
- The routing table is stable with minimal changes as the protocol does not react to issues within the Autonomous system.
- EGP is a simple reachability protocol, and, unlike modern distance-vector and path-vector protocols, it is limited to tree-like topologies and does not support multipath networking environments, making it less efficient.
- Since this routing protocol is designed to be centrally controlled, it reduces the scalability which is a major drawback in today's fast growing Internet.
- Another reason is that no central authority controls the commercialized Internet. The Internet is composed of many equal networks.

Border Gateway Protocol (BGP)

Border Gateway Protocol (BGP) is used to Exchange routing information for the internet and is the protocol used between ISP which are different Autonomous systems.

What is an autonomous system?

- The Internet is a network of networks, and autonomous systems are the big networks that make up the Internet.
- More specifically, an autonomous system (AS) is a large network or group of networks that has a unified routing policy.
- Every computer or device that connects to the Internet is connected to an AS.



➤ The protocol can connect together any internetwork of autonomous system using an arbitrary topology.

➤ The only requirement is that each AS have at least one router that is able to run BGP and that is router connect to at least one other AS's BGP router.

➤ BGP's main function is to exchange network reach-ability information with other BGP systems.

➤ Border Gateway Protocol constructs an autonomous systems' graph based on the information exchanged between BGP routers.

Characteristics of Border Gateway Protocol (BGP):

Inter-Autonomous System Configuration: The main role of BGP is to provide communication between two autonomous systems.

BGP supports Next-Hop Paradigm.

Coordination among multiple BGP speakers within the AS (Autonomous System).

Path Information: BGP advertisement also include path information, along with the reachable destination and next destination pair.

Policy Support: BGP can implement policies that can be configured by the administrator. For ex:- a router running BGP can be configured to distinguish between the routes that are known within the AS and that which are known from outside the AS.

Runs Over TCP.

BGP conserve network Bandwidth.

BGP supports CIDR.

BGP also supports Security.

Functionality of Border Gateway Protocol (BGP):

BGP peers performs 3 functions, which are given below.

- The first function consist of initial peer acquisition and authentication. both the peers established a TCP connection and perform message exchange that guarantees both sides have agreed to communicate.
- The second function mainly focus on sending of negative or positive reach-ability information.
- The third function verifies that the peers and the network connection between them are functioning correctly.

BGP Route Information Management Functions:

Route Storage:

Each BGP stores information about how to reach other networks.

Route Update:

In this task, Special techniques are used to determine when and how to use the information received from peers to properly update the routes.

Route Selection:

Each BGP uses the information in its route databases to select good routes to each network on the internet network.

Route advertisement:

Each BGP speaker regularly tells its peer what it knows about various networks and methods to reach them.

Difference between EIGRP and BGP

SR.NO	EIGRP	BGP
1.	EIGRP stands for Enhanced Interior Gateway Routing Protocol.	BGP stands for Border Gateway Protocol.
2.	It works on Dual distance vector algorithm.	It works on Best path algorithm.
3.	It is basically use for lager size organization.	It is basically use for very lager size organization as compared to EIGRP.
4.	It is hybrid type.	It is a Vector State type.
5.	It is interior gateway protocol.	It is external gateway protocol.
6.	It is a intelligent routing protocol.	It is a more intelligent routing protocol than EIGRP.
7.	In EIGRP, Internet protocol is used.	In BGP, Transmission control protocol is used.
8.	Its administrative distance are 90 (Internal) and 170 (External).	Its administrative distance are 20 (Internal) and 200 (External).